

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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Unit I | Session 4 | SLO 2: Paradigm Hierarchy & Features (Code: 1042)

Q: 1. Conceptual: Arrange the following in a correct hierarchy:

• Imperative • Structured • Procedural • Object-Oriented

Ans: Correct Architectural Hierarchy:

1. **Imperative Programming** (Broadest category: state-manipulation paradigm)
2. ■■■ **Structured Programming** (Sub-discipline: eliminates goto using sequence, selection, iteration)
3. ■■■ **Procedural Programming** (Modular extension: groups instructions into routines/procedures)
4. ■■■ **Object-Oriented Programming** (Modern evolution: bundles state and behavior into encapsulated objects)

Q: 2. Definition Match: Match each paradigm to its defining feature:

• Functional • Object-Oriented • Procedural • Logic-Based

- a) Uses facts and rules for computation
- b) Focuses on objects and classes
- c) Uses functions to avoid mutable state
- d) Breaks programs into procedures or routines

Ans: Matches:

- **Functional** → c) Uses functions to avoid mutable state
- **Object-Oriented** → b) Focuses on objects and classes
- **Procedural** → d) Breaks programs into procedures or routines
- **Logic-Based** → a) Uses facts and rules for computation

Q: 3. True or False:

- a) All object-oriented programs are procedural.
- b) Structured programming is a subset of imperative programming.
- c) Functional programming encourages global variables.

Ans: Answers:

- a) **False:** OOP encapsulates state inside objects rather than relying on top-level procedural decomposition.
- b) **True:** Structured programming is a formalized subset of imperative programming that restricts control flow.
- c) **False:** Functional programming strictly avoids mutable state and global variables to prevent side effects.